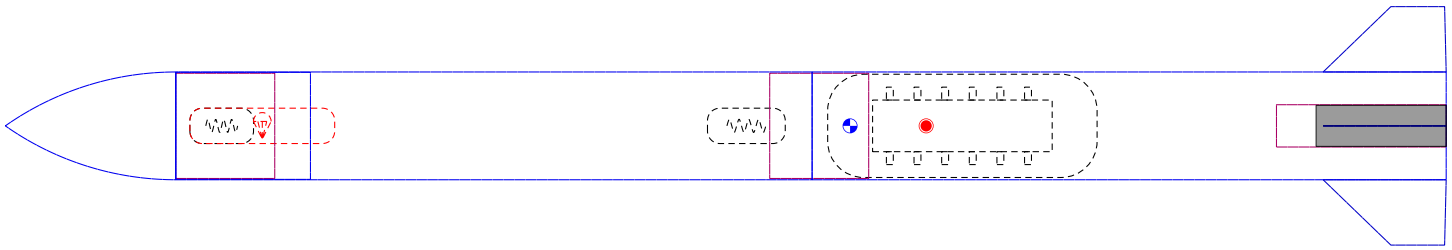
Rocket Design



BLEP

Stages: 1

Mass (with motor): 706 g

Stability: 0.706 cal / 5.28 %

CG: 59.7 cm

CP: 65.1 cm

F10-8

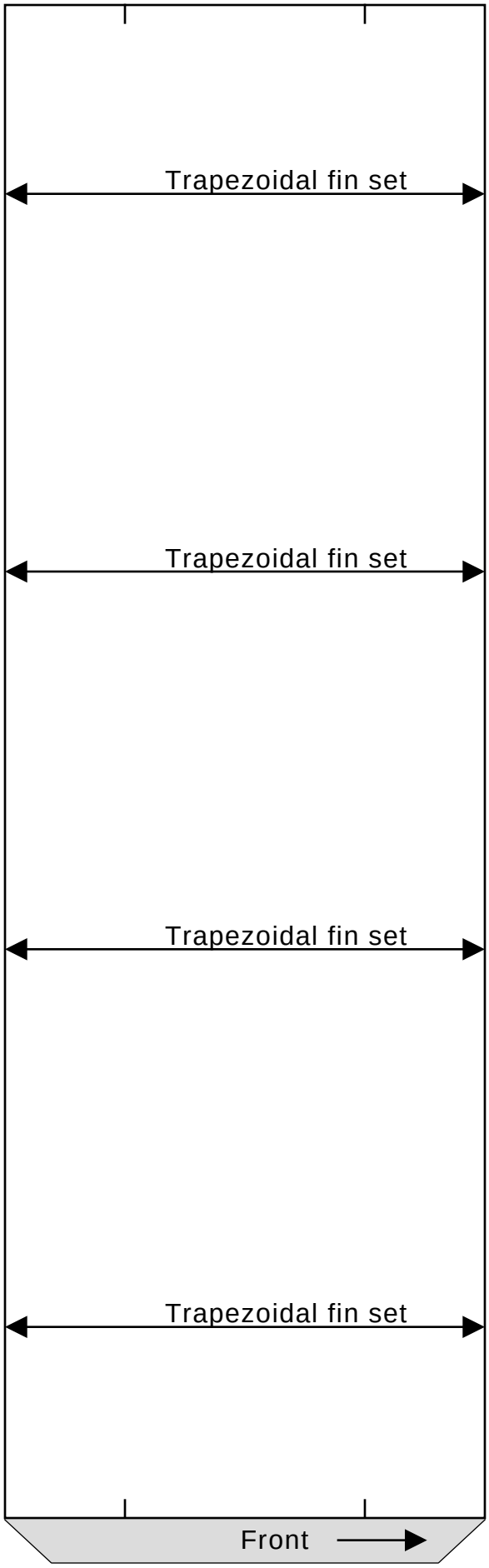
Altitude	207 m
Flight Time	25.7 s
Time to Apogee	9.6 s
Optimum Delay	2.52 s
Velocity off Pad	7.37 m/s
Max Velocity	32.2 m/s
Velocity at Deployment	11.1 m/s
Landing Velocity	13.7 m/s

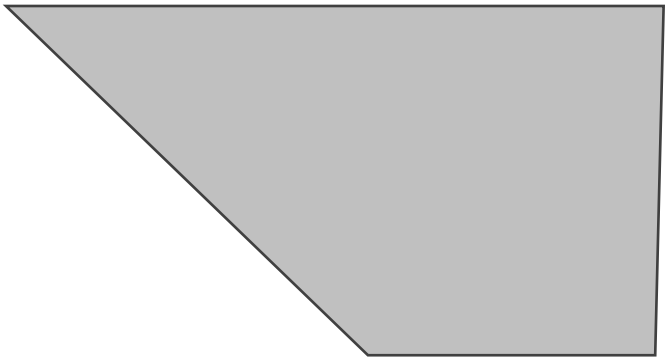
Motor	Avg Thrust	Burn Time	Max Thrust	Total Impulse	Thrust to Wt	Motor Wt	Size
F10	10.9 N	7.02 s	28.2 N	76.3 Ns	1.57:1	40.7 g	29/92 mm

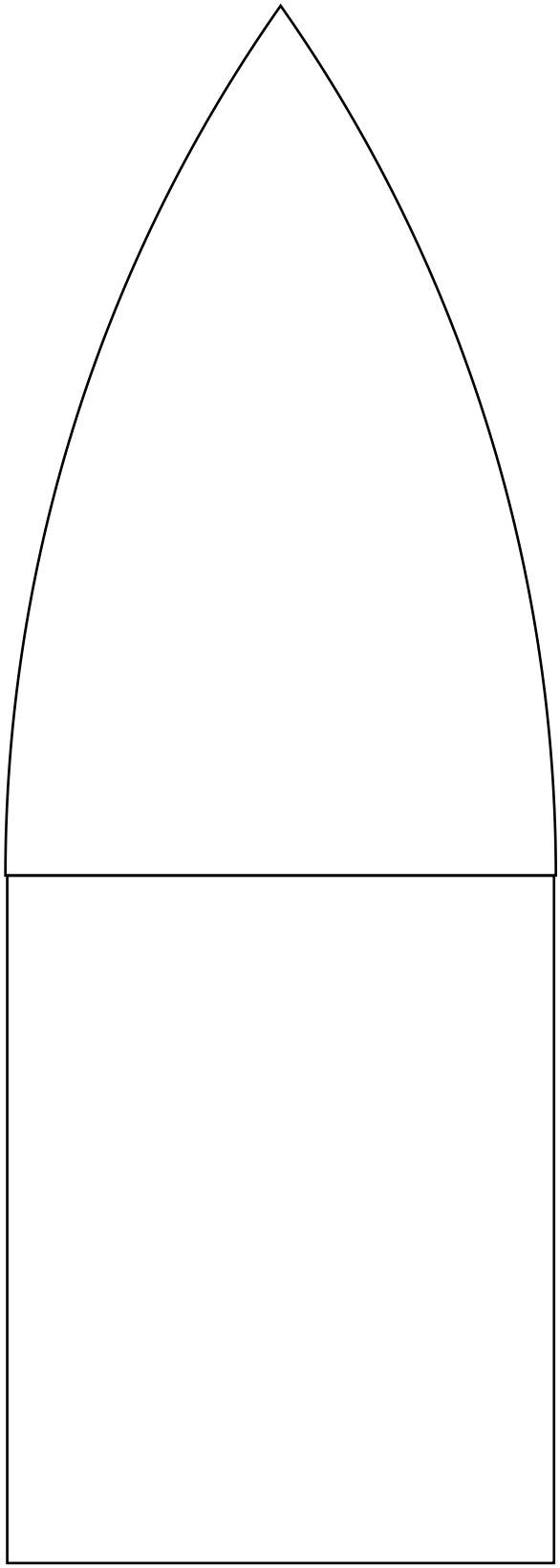
Parts Detail

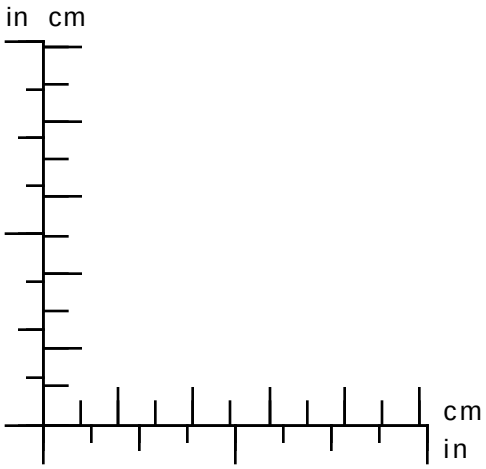
Stage: Sustainer

	Nose cone	[material:[material:polystyrene PS]] (1 g/cm³)	Ogive	Len: 12.1 cm	Mass: 200 g
	Body tube	Cardboard (0.68 g/cm³)	Dia_{in} 7.44 cm Dia_{out} 7.62 cm	Len: 45 cm	Mass: 65.1 g
	Tube coupler	Cardboard (0.68 g/cm³)	Dia_{in} 7.44 cm Dia_{out} 7.44 cm	Len: 7 cm	Mass: 0 g
	Parachute	Ripstop nylon (67 g/m²)	Dia_{out} 30 cm	Len: 10.2 cm	Mass: 7.98 g
	Shroud Lines	Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)	Lines: 6	Len: 30 cm	
	Shock cord	Kevlar thread 800 (1.1 mm, 3/64 in) (0.992 g/m)		Len: 40 cm	Mass: 0.397 g
	Body tube	Cardboard (0.68 g/cm³)	Dia_{in} 7.44 cm Dia_{out} 7.62 cm	Len: 44.8 cm	Mass: 64.9 g
	Tube coupler	Cardboard (0.68 g/cm³)	Dia_{in} 7.44 cm Dia_{out} 7.44 cm	Len: 7 cm	Mass: 0 g
	Flight Computer		Dia_{out} 7.3 cm		Mass: 96 g
	Inner Tube	Cardboard (0.68 g/cm³)	Dia_{in} 2.9 cm Dia_{out} 2.99 cm	Len: 12 cm	Mass: 3.4 g
	Trapezoidal fin set (4)	PLA - 100% infill (1.25 g/cm³)	Thick: 1.27 cm		Mass: 183 g
	Shock cord	Kevlar thread 800 (1.1 mm, 3/64 in) (0.992 g/m)		Len: 40 cm	Mass: 0.397 g


Trapezoidal fin set
Trapezoidal fin set
Trapezoidal fin set
Trapezoidal fin set
Front



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